

01/02/26

Aim:-

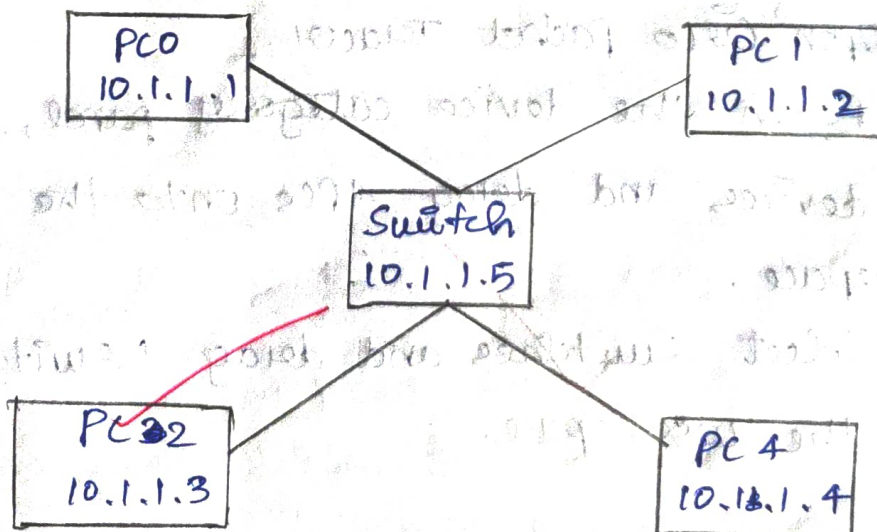
To design, set up, and configure a Local Area Network (LAN) connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource/data sharing across the network.

Equipment Required:-

S.No	Equipment	Quantity	Specification
1.	Personal computer (PC)	4	Ethernet (NIC) enabled
2.	Network switch	1	8/16/24 port, unmanaged/managed
3.	Ethernet cables	4	Straight-through, RJ-45 connectors

Network diagram:-

Star topology LAN



IP Address Configuration Table

Device	IP Address	Subnet Mask	Connected to
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
Switch (management)	10.1.1.5	255.0.0.0	VLAN 1 Interface

Implementation In Cisco Packet Tracer

Since physical hardware was limited, the LAN was also implemented and verified using Cisco Packet Tracer simulation software.

Step 1: Place the devices

1. Open Cisco Packet Tracer
2. From the device category panel, select End devices and drag 4 PCs onto the workspace.
3. Select ~~switches~~ and drag 1 switch onto the workspace.

Step 2: Cable the devices

4. Click Connections → select copper, Straight-through cable.
5. Click each PC, then click the switch, and select an available FastEthernet port to complete ~~each~~ each connection.
6. Repeat for all four PCs. Green Link lights confirm a successful physical connection.

Step 3: Assign IP Addresses to PC

7. Click a PC → Desktop tab → IP Configuration.
8. Enter the Static IP address and Subnet Mask as per the IP table for each PC.

Step 4: Configure the switch management IP

9. Click the Switch → CLI tab.
10. Enter the following commands to assign a management IP:

enable

configure terminal

interface VLAN1

ip address 10.1.1.5 255.0.0.0

no shutdown

exit

Step 5: Verify Connectivity using ping

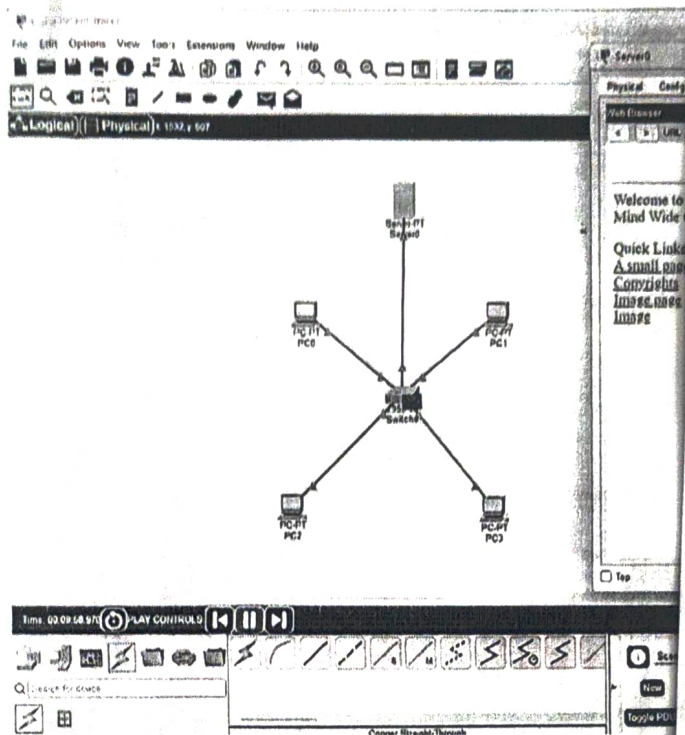
11. Click a PC → Desktop tab → Command Prompt
12. Use the ping command to test connectivity to the other PCs eg: ping 10.1.1.2
13. Successful "Reply from" messages confirm connectivity.

Step 6: Demonstrate Data Sharing

14. Add a server device and connect it to the switch.
15. Assign it an IP address via its Desktop → IP configuration table.
16. Open the server's Services tab and enable the HTTP service.
17. From another PC, open the web browser and enter the server's IP address to load the page, confirming successful data transfer across the LAN.

Step 7: Observe Packet flow in Simulation mode

18. Switch to Simulation mode
19. Send a ping or web request between two devices and use the Play/Capture Controls to step through the switch, observing MAC address based forwarding.



After clicking on the 'Run' button, the simulation starts. The network topology is displayed, showing a central switch connected to four PCs (PC0, PC1, PC2, PC3). The simulation shows the data flow between the PCs and the switch. The status bar at the bottom indicates the simulation is running.

Result:-
 This setting up LAN was executed successfully.